Problem 17

Prove that the events

$$A,\ \bar{A}B,\ \text{and}\ \overline{A+B}$$

form a complete system of events (partition of the sample space).

Solution

A complete system of events (or partition) requires that the events are:

1. **Mutually exclusive** (pairwise disjoint): No two events can occur simultaneously
2. **Exhaustive**: Their union equals the entire sample space S

Step 1: Simplify $\overline{A+B}$

Using De Morgan's law:

$$\overline{A+B} = \bar{A}\bar{B}$$

So we need to prove that A, $\bar{A}B$, and $\bar{A}\bar{B}$ form a complete system.

Step 2: Prove Mutual Exclusivity (Pairwise Disjoint)

Check $A \cap \bar{A}B$

$$A \cap \bar{A}B = A \cdot \bar{A} \cdot B = \emptyset \cdot B = \emptyset$$

Since $A\bar{A} = \emptyset$, the events A and $\bar{A}B$ are disjoint.

Check $A \cap \bar{A}\bar{B}$

$$A \cap \bar{A}\bar{B} = A \cdot \bar{A} \cdot \bar{B} = \emptyset \cdot \bar{B} = \emptyset$$

The events A and $\bar{A}\bar{B}$ are disjoint.

Check $\bar{A}B \cap \bar{A}\bar{B}$

$$\bar{A}B \cap \bar{A}\bar{B} = \bar{A} \cdot B \cdot \bar{A} \cdot \bar{B} = \bar{A}^2 \cdot B\bar{B}$$

Since $B\bar{B} = \emptyset$:

$$= \bar{A} \cdot \emptyset = \emptyset$$

The events $\bar{A}B$ and $\bar{A}\bar{B}$ are disjoint.
Conclusion: All three events are pairwise mutually exclusive.

Step 3: Prove Exhaustivity (Union Equals Sample Space)

We need to show:

$$A + \bar{A}B + \bar{A}\bar{B} = S$$

Simplify the Union

$$A + \bar{A}B + \bar{A}\bar{B}$$

Factor $\bar{A}$ from the last two terms:

$$= A + \bar{A}(B + \bar{B})$$

Using the complement law: $B + \bar{B} = S$

$$= A + \bar{A} \cdot S = A + \bar{A}$$

Using the complement law: $A + \bar{A} = S$

$$= S$$

Therefore, the union of the three events equals the entire sample space.

Step 4: Verification Using Set Interpretation

Let's interpret these events in terms of sets:

- A: Event A occurs, B may or may not occur
- $\bar{A}B$: Event A does not occur, but B occurs
- $\bar{A}\bar{B}$: Neither A nor B occurs

These three cases are:

1. A occurs (regardless of B)
2. A doesn't occur, but B does
3. Neither A nor B occurs

These cases are clearly exhaustive and mutually exclusive, covering all possible outcomes.

Step 5: Visual Representation (Venn Diagram)

In a Venn diagram with circles for A and B:

- A consists of all points in circle A (including $A \cap B$ and $A \cap \bar{B}$)
- $\bar{A}B$ is the part of circle B outside circle A
- $\bar{A}\bar{B}$ is the region outside both circles

These three regions partition the entire sample space with no overlap.

Alternative Proof: Using Boolean Algebra

We can also express any outcome in the sample space in terms of these events.

Any element $\omega \in S$ satisfies exactly one of:

- $\omega \in A$ (then ω is in the first event)
- $\omega \in \bar{A}$ and $\omega \in B$ (then ω is in the second event $\bar{A}B$)
- $\omega \in \bar{A}$ and $\omega \in \bar{B}$ (then ω is in the third event $\bar{A}\bar{B}$)

Since every element must either be in A or not ($\bar{A}$), and if in $\bar{A}$ must either be in B or not ($\bar{B}$), we have covered all possibilities.

Probabilistic Verification

If we assign probabilities:

$$P(A) + P(\bar{A}B) + P(\bar{A}\bar{B}) = P(S) = 1$$

This confirms that the events form a partition.
Also:

$$P(A \cap \bar{A}B) = P(A \cap \bar{A}\bar{B}) = P(\bar{A}B \cap \bar{A}\bar{B}) = 0$$

confirming mutual exclusivity.

Final Answer

$$\boxed{\text{The events } A, \bar{A}B, \text{ and } \overline{A+B} \text{ form a complete system of events}}$$

Proof summary:
1. Mutual Exclusivity:

$$\begin{aligned} A \cap \bar{A}B &= \emptyset \\ A \cap \bar{A}\bar{B} &= \emptyset \\ \bar{A}B \cap \bar{A}\bar{B} &= \emptyset \end{aligned}$$

2. Exhaustivity:

$$\begin{aligned} A + \bar{A}B + \bar{A}\bar{B} &= A + \bar{A}(B + \bar{B}) \\ &= A + \bar{A} \\ &= S \end{aligned}$$

Interpretation: These three events partition the sample space into:

- Event A occurs (with or without B)
- Event A doesn't occur, but B does
- Neither A nor B occurs

This covers all possibilities exactly once, forming a complete partition.